EXPERIMENT – 12

PORT SCANNING TECHNIQUES

AIM

To execute different port scanning techniques using Nmap in Kali Linux and observe the scan results.

DESCRIPTION

Nmap is a network scanning tool used to examine ports on a target host. This experiment demonstrates four techniques: TCP SYN scan, TCP connect scan, UDP scan, and TCP ACK scan. Run these commands only against a system you own or have permission to test.

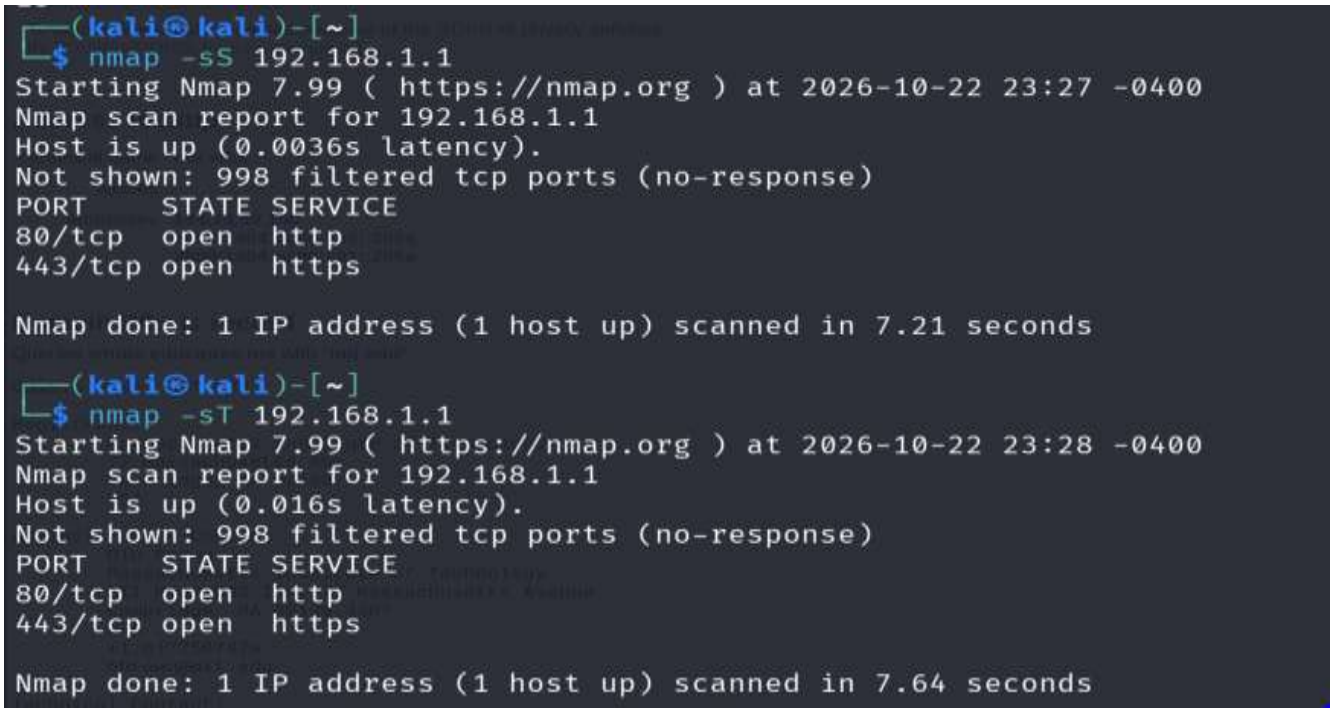
SCANNING TECHNIQUES

Flag	Use	Example
-sS	TCP SYN port scan	nmap -sS 192.168.1.1
-sT	TCP connect port scan	nmap -sT 192.168.1.1
-sU	UDP port scan	nmap -sU 192.168.1.1
-sA	TCP ACK port scan	nmap -sA 192.168.1.1

PROCEDURE

1. Start Kali Linux and open the Terminal.
2. Replace 192.168.1.1 with the IP address of an authorized target system on your lab network.
3. Run the TCP SYN scan command: nmap -sS 192.168.1.1.
4. Run the TCP connect scan command: nmap -sT 192.168.1.1.
5. Run the UDP scan command: nmap -sU 192.168.1.1.
6. Run the TCP ACK scan command: nmap -sA 192.168.1.1.
7. Observe and record the port states and scan summaries displayed in the terminal.

OUTPUT





RESULT

The TCP SYN, TCP connect, UDP, and TCP ACK scanning techniques were executed using Nmap, and the resulting port information was observed.